



FIRST SEMESTER: 2026-2027
COURSE HANDOUT

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : PHY U102
Course Title : **Introduction to Classical and Quantum Waves**
Instructor-in-Charge : **Madhukar Mishra**

1. Course Description: Introduction to oscillations, simple harmonic motion, superposition of harmonic oscillations, damped and forced oscillations, resonance and quality factor: transverse and longitudinal waves, wave speed in a stretched string, wave equation in one dimension, traveling solutions to the wave equation, energy transport in waves, standing waves, travelling sound waves; Interference of light waves, introduction to diffraction; concept of photon, photoelectric effect, Compton scattering; wave-particle duality, de Broglie hypothesis and matter waves; 1D Quantum wave functions: properties and applications; Pauli's exclusion principle and multielectrons atoms, introduction to X-rays and Lasers.

2. Scope & Objective of the Course: This course introduces the fundamental concepts of classical and quantum waves, beginning with simple harmonic motion as the basis of oscillatory phenomena and progressing through mechanical waves, sound, interference, and diffraction. It then explores the transition from classical to quantum physics by examining the wave-particle nature of light and matter, culminating in an elementary introduction to quantum wave functions and their applications to atomic systems, X-rays, and lasers. Designed for undergraduate students with a basic knowledge of calculus, the course emphasizes physical intuition, conceptual understanding, and qualitative reasoning over advanced mathematical formalism. Its objective is to provide a unified understanding of wave phenomena across classical and quantum physics, establish the conceptual foundations of modern physics, and prepare students for advanced courses in optics, quantum mechanics, condensed matter physics, and related disciplines.

3. Textbooks:

(i) Principles of Physics (10th Edition) by Halliday, Resnick, and Walker, John Wiley & Sons. **(T.B.)**

4. Reference Books:

(i) Physics, Vols. 1 & 2, by Halliday, Resnick, & Krane, 5th Ed., John Wiley & Sons. 2002.

(ii) Berkeley Physics Course, Vol. 3, Waves, by Frank S. Crawford Jr, Special Indian Edition, McGraw-Hill, 2011.

(iii) Fundamentals of Optics, Fourth International Edition, Jenkins & White, McGraw-Hill, 1981.

(iv) Concepts of Modern Physics, 7th Edition, Arthur Beiser, McGraw-Hill, 2017

5. Course Learning Outcomes:

(i) Explain the principles of simple harmonic motion and analyze damped and forced oscillations, resonance, and the superposition of oscillatory motion as the foundation of wave phenomena.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

(ii) Describe the behavior of classical waves, including wave propagation, wave equations, energy transport, sound waves, and the formation of standing waves in one-dimensional systems.

(iii) Explain the wave nature of light through interference and diffraction and discuss the experimental observations that established the quantum nature of light, including the photoelectric effect and Compton scattering.

(iv) Describe the concept of matter waves by explaining wave-particle duality, the de Broglie hypothesis, and the extension of wave concepts from light to material particles.

(v) Interpret the physical meaning of quantum wave functions in one dimension and explain their role in describing quantum systems, including an introductory understanding of multi-electron atoms through Pauli's exclusion principle and the principles of X-rays and lasers.

6. Course Plan:

Module No.	Lecture No.	Reference	Learning outcomes
1. SHM	1-2	T.B. 15.1	- Simple Harmonic Motion (S.H.M), Definition of various parameters of S.H.M. - Kinetic and potential energies of S.H.M. - Angular SHM, Pendulum, Circular motion - Damped and Forced oscillations, and phenomenon of resonance
	3	T.B. 15.2	
	4-5	T.B. 15.3-15.4	
	6	T.B. 15.5-15.6	
2. Waves	7	T.B. 16.1	- Transverse & longitudinal waves including different parameters related to waves - Relationship between wave speed, tension and linear density of a stretched string. - Energy and power of a traveling wave through string - Principle of superposition for waves, Standing waves
	8	T.B. 16.2	
	9	T.B. 16.3	
	10	T.B. 16.5, 16.7	
3. Light Waves	11	T.B. 35.1	- Light as a wave - Young's interference experiment (In)Coherent light, double slit interference - Interference from thin film - Single slit diffraction - Double slit diffraction
	12	T.B. 35.2	
	13	T.B. 35.3	
	14-15	T.B. 35.4	
	16-17	T.B. 36.1-36.2	
	18	T.B. 36.4	
4. Matter Waves	18	T.B. 38.1	- Concept of photon - Photoelectric effect - Compton scattering - Birth of quantum physics, Wave-particle duality - Schroedinger equation, its application in 1D problems
	19	T.B. 38.2	
	20	T.B. 38.3	
	21-22	T.B. 38.4, 38.5	
	23-24	T.B. 38.6, 38.8-38.9	
5. Wave function in Quantum Physics	23	T.B. 38.8	- Reflection from a potential step - Tunneling through a potential barrier - Wave functions and energies of a trapped electron - Two and three dimensional electron traps - Pauli's exclusion principle and many electron atoms
	24	T.B. 38.9	
	25-26	T.B. 39.1-39.2	
	27	T.B. 39.4	
	28	T.B. 40.4	

8. Evaluation Scheme:

Components	Duration	Marks (%)	Date & Time	Nature of component
------------	----------	-----------	-------------	---------------------



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

				(Closed Book/ Open Book)
Mid-Semester test	90 mins.	30%	07/10 AN2	Closed Book
Tutorial Tests (Best 3 out of 4)	30 mins.	30%	To be announced	Closed Book
Comprehensive Examination	180 mins.	40%	10/12 AN	Closed + Open Book

- **Attendance Policy:** Each student is expected to attend all classes, lab sessions, and participate in all evaluation components.
- **NC Criteria:** 30% of the average of the top three performers or 40% of the median, whichever is lower.

9. Chamber Consultation Hour: Will be announced in the class. Also, feel free to email me to seek a prior appointment.

10. Notices: On Google Classroom.

11. Make-up Policy: Stringent. Only in the most genuine cases, as judged by the IC. Make-up will not be given for the tutorial tests.

Instructor-in-charge
Course No. PHY U102